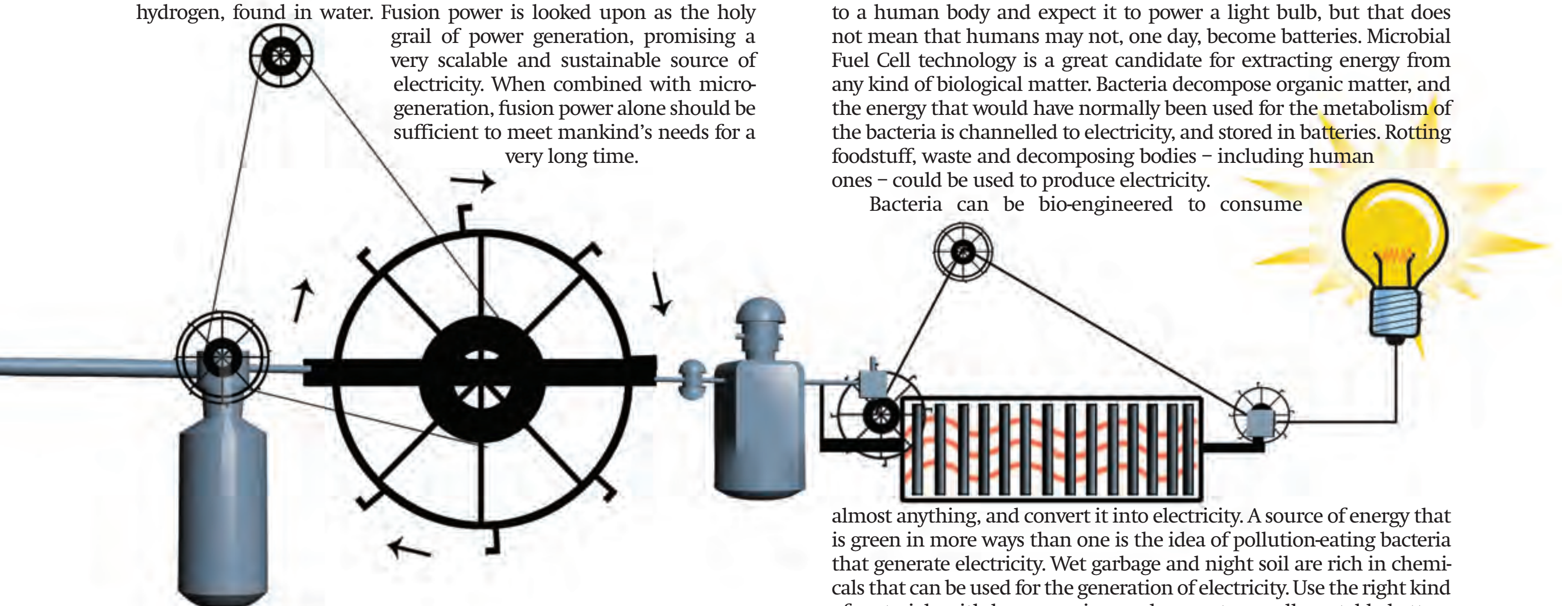
with fusion power for a long time now, but the technology is proving surprisingly difficult to develop. This is mostly because when breakthroughs occur, fresh obstacles show up. The biggest problem lies in containing the very hot plasma required for fusion to take place, for which electrostatic or magnetic fields are currently used. The raw materials for fusion are deuterium and tritium, heavy isotopes of hydrogen, found in water. Fusion power is looked upon as the holy grail of power generation, promising a very scalable and sustainable source of electricity. When combined with microgeneration, fusion power alone should be sufficient to meet mankind's needs for a very long time.



Recycling energy

The current methods of production and consumption of energy are very wasteful. A large chunk of the energy is not used for the final work, or gets dissipated as heat. A lot of electricity use is terribly inefficient. For example, power plants consume fossil fuel to boil water. The steam turns turbines that then produce electricity. This electricity is then converted back into heat to warm up a home, or prepare warm water. A solar water heater would reduce the maintenance cost of a grid, and conserve the energy lost in the various stages of production, transmission and consumption. Scientists will look to install usage-specific sources of electricity. Street lights and traffic signals around the world have already sprouted solar panels. The next step in this direction is to research innovative ways to convert wasted energy into electricity.

Some kinds of materials generate electricity when put under pressure. Metal plates beneath roads could theoretically be used to power up the streetlights. Electricity could be stored during the daytime, and used at the night. Similarly, plates could be placed below mass transport systems such as trains. Although generating enough electricity to power the trains themselves is impossible according to the laws of physics, auxiliary devices such as signal lights, fans, lights, indicators and doors could be powered by the movement of the trains. This would require more energy to drive the train, but overall could be more efficient. Existing vehicles do use some of the kinetic energy for some of the electric needs. The Toyota Prius for example is a hybrid car that does just this. At smaller levels, certain crystals produce electricity when pressure is applied. This principle could be used to power gadgets; a gaming console, mobile phone or computer could be fed back at least some of the required power from the effort of pressing the keys.

Converting the energy from the movement of the human body into power with portable instruments is an idea that has been around for a long time. Kinetic rotors power watches, and some of them even power personal gadgets (refer to Drool Maal in this issue). Torches, radios, mobile phones, and cameras are all ideal candidates for using the kinetic energy released from human movement. The technology for this already exists but costs more to mass produce and deploy than traditional battery-driven gadgets. An MP3 player that juices up as you jog is a great idea, and there would be many takers for such a gadget. The California Fitness Center in Hong Kong is a gym that draws power from the patrons who work out, and use it to light up the place. In the future, expect to see more such usage specific ways to draw power.

Enter the Matrix

Groundbreaking research suggests humans, potatoes, trees, cows and bacteria as possible sources of electricity. You can't just connect wires to a human body and expect it to power a light bulb, but that does not mean that humans may not, one day, become batteries. Microbial Fuel Cell technology is a great candidate for extracting energy from any kind of biological matter. Bacteria decompose organic matter, and the energy that would have normally been used for the metabolism of the bacteria is channelled to electricity, and stored in batteries. Rotting foodstuff, waste and decomposing bodies – including human ones – could be used to produce electricity.

Bacteria can be bio-engineered to consume almost anything, and convert it into electricity. A source of energy that is green in more ways than one is the idea of pollution-eating bacteria that generate electricity. Wet garbage and night soil are rich in chemicals that can be used for the generation of electricity. Use the right kind of materials with human urine, and you get a small, portable battery that lasts a few days. The digestive juices of cattle have also been used to generate electricity. Bacteria can be used to decompose human and livestock waste, and then the waste itself is reduced to a form of ethanol, which is then used as fuel in power plants. This is not a new idea, but an idea that has not been implemented for a long time. As much as a century ago, gas from rotting vegetables was used to light street lamps. Another idea applies pressure and heat to biological waste, mimicking the natural processes that many assume produced crude oil. Although the technology exists, the efficiency is not perfect, and development will take some time.

Just connect wires to a potato or a lemon, and surprisingly, it will power a (small) light bulb. You will have to find a very low power light-bulb though. Large potato farms as a source of energy have been around in popular culture for some time now. While this idea may not be viable, there are similar ideas that can be put into practice in a practical way. For example, trees can line streets and be used to power the street lamps.

Supernova Batteries

Some of the ideas that the scientific community has are beyond the scope of current technology. The Sun, for example, is a great source of energy. The incident solar radiation on a small part of the surface of the Earth, is a viable option. A modern industrial nation receives solar energy equivalent to approximately 1,000 times its normal energy usage. Imagine large solar arrays in space that beam back energy in the form of microwaves to Earth (killing countless birds — ed.). Now, imagine a ring of solar arrays arranged all around the sun, and the amount of energy it could beam back to the planet. How about a star completely enclosed within a spherical solar panel — you have what they call a Dyson sphere. This kind of a structure would capture all the power released from a star.

Exotic materials or particles such as miniature black holes and anti-matter are thought of as great sources of energy. Anti-matter has been created, and power extracted too, but the particles are very rare and very costly to be commercially viable at present. However, there is active research going on in particle physics, and there may be properties or particles that are hidden from human science at this point of time. ■

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